In [4]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load your CSV file
df = pd.read_csv("Titanic-Dataset.csv")

# Display first 5 rows
print(df.head())

# Dataset information
print("\nDataset Shape:")
print(df.shape)

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())
```

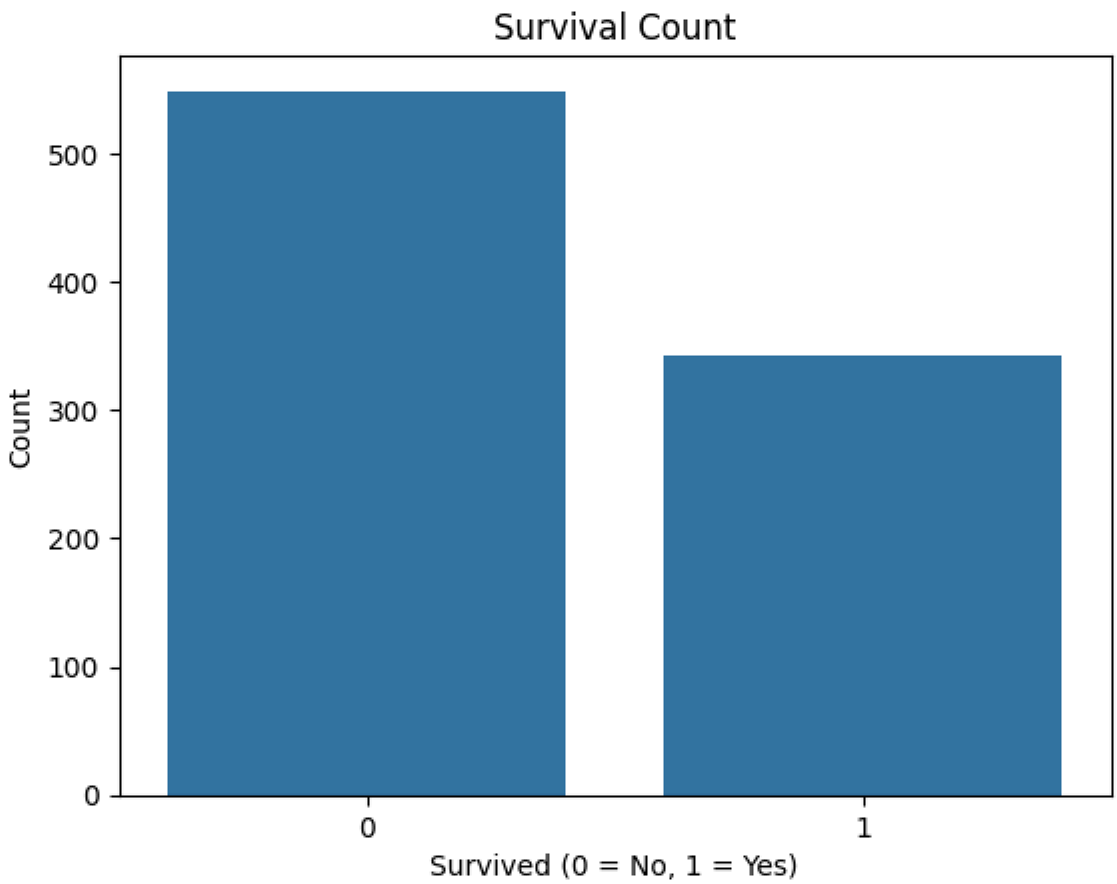
```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S

Dataset Shape:
(891, 12)

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

Missing Values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
```

```
Cabin          687
Embarked         2
dtype: int64
```

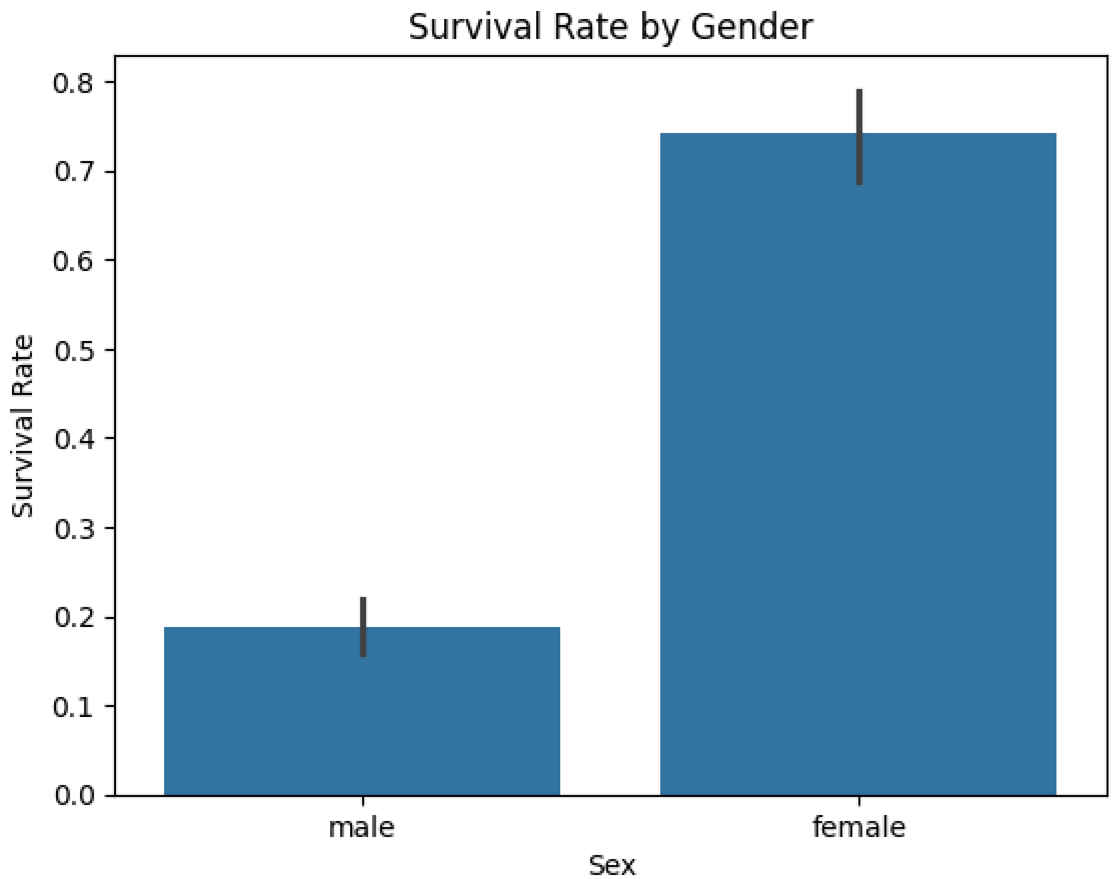
In [8]:
```
df = df.drop_duplicates()

print("\nShape after removing duplicates:")
print(df.shape)

# Fill missing Age with median
df["Age"] = df["Age"].fillna(df["Age"].median())

# Fill missing Embarked with mode
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Cabin has many missing values, so remove it
df = df.drop(columns=["Cabin"])

print("\nMissing Values After Cleaning:")
print(df.isnull().sum())

print("\nData Types:")
print(df.dtypes)

print("\nFinal Dataset Shape:")
print(df.shape)

print("\nCleaned Dataset:")
print(df.head())
```

```
Shape after removing duplicates:
(891, 12)

Missing Values After Cleaning:
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64

Data Types:
PassengerId      int64
Survived         int64
Pclass           int64
Name               str
Sex                str
Age            float64
SibSp            int64
Parch            int64
Ticket             str
Fare           float64
Embarked           str
dtype: object

Final Dataset Shape:
(891, 11)

Cleaned Dataset:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Embarked
0      0         A/5 21171   7.2500        S
1      0          PC 17599  71.2833        C
2      0  STON/O2. 3101282   7.9250        S
3      0            113803  53.1000        S
4      0            373450   8.0500        S
```

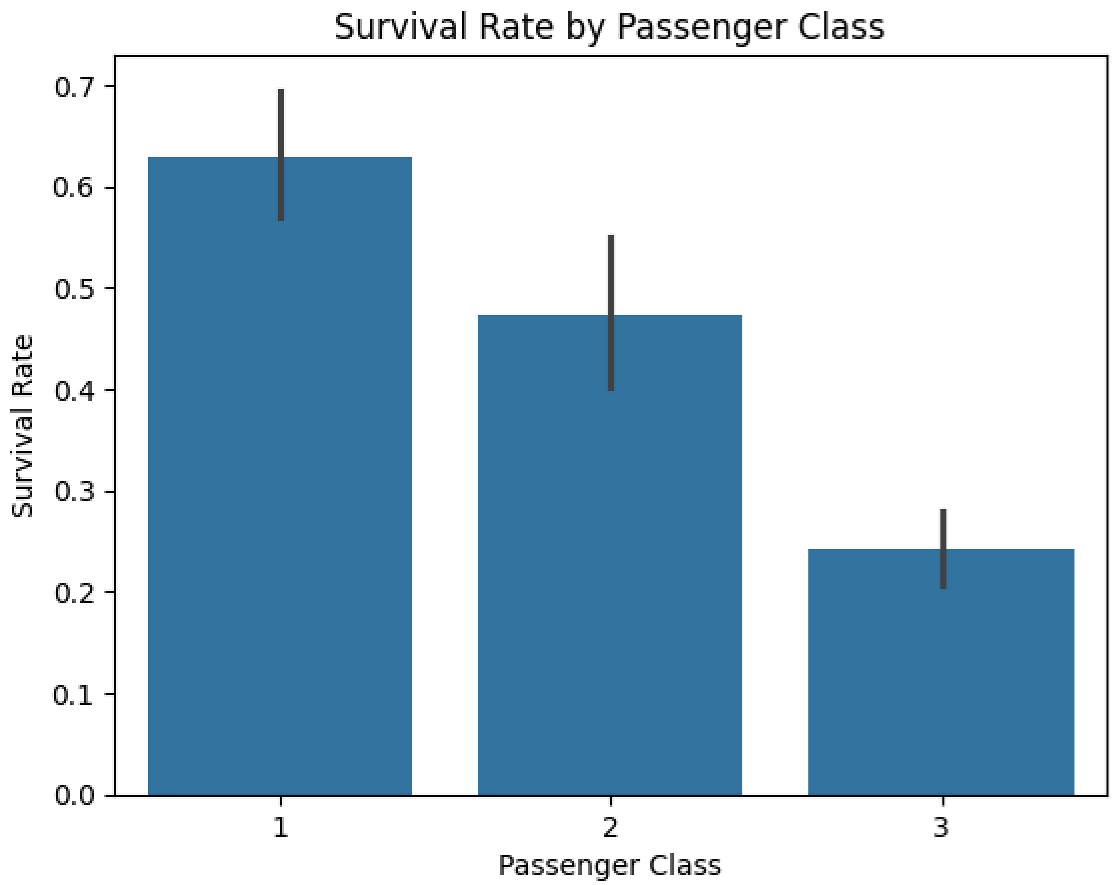
In [9]:
```
# Survival Count
sns.countplot(data=df, x="Survived")
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```



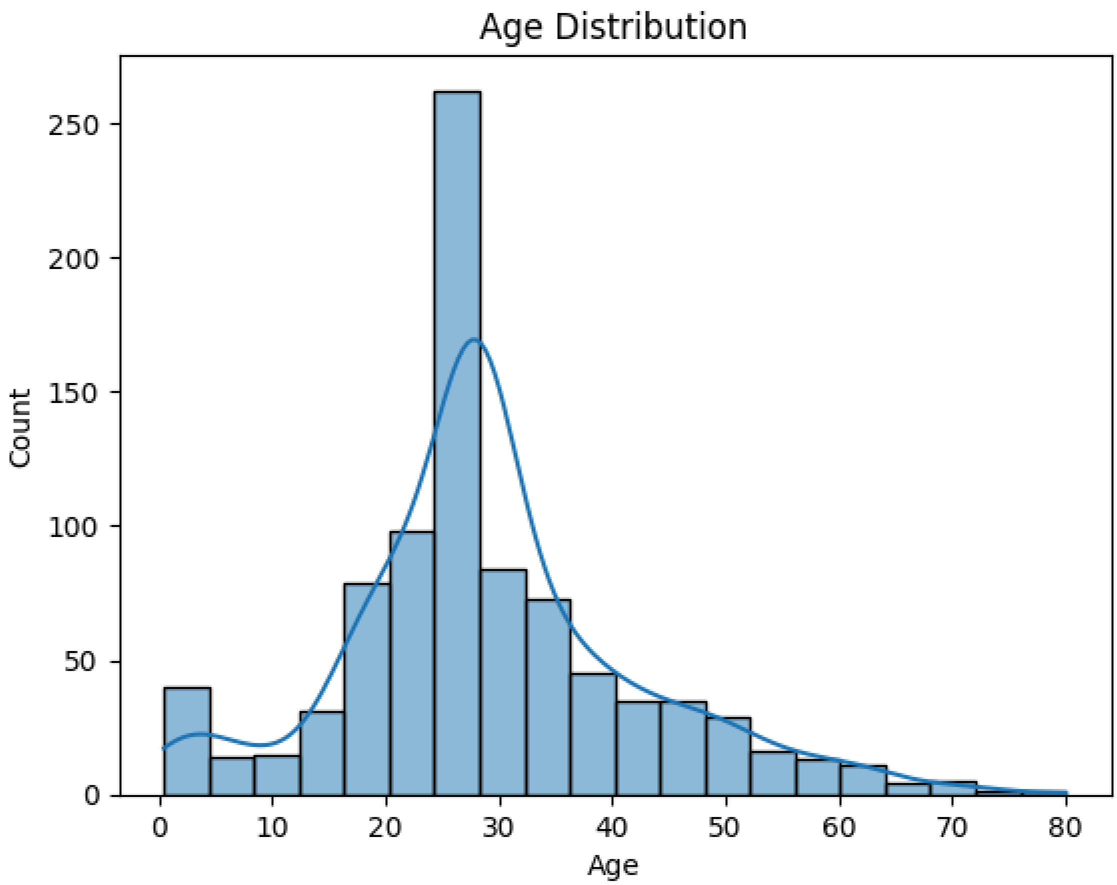
In [11]:
```
# Survival by Gender
sns.barplot(data=df, x="Sex", y="Survived")
plt.title("Survival Rate by Gender")
plt.ylabel("Survival Rate")
plt.show()
```



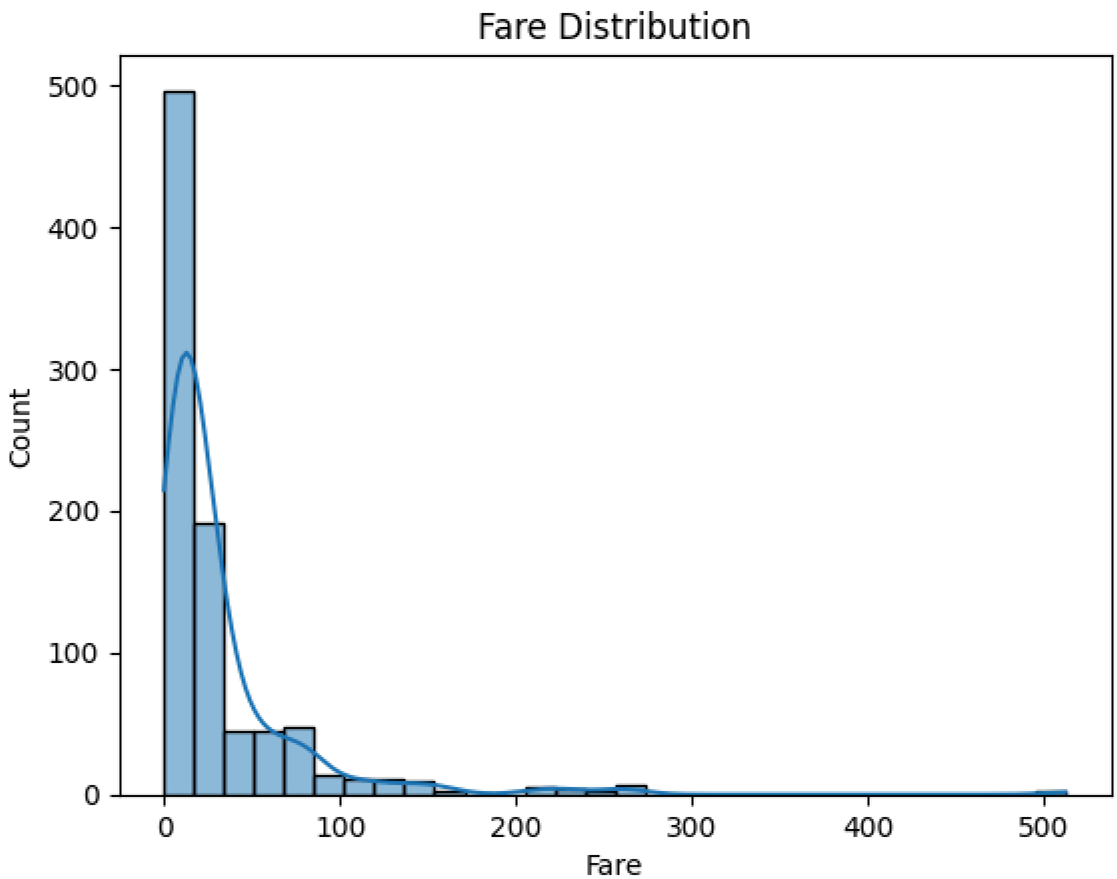
In [12]:
```
# Survival by Passenger Class
sns.barplot(data=df, x="Pclass", y="Survived")
plt.title("Survival Rate by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Survival Rate")
plt.show()
```



In [13]:
```
# Age Distribution
sns.histplot(data=df, x="Age", bins=20, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



In [14]:
```
# Fare Distribution
sns.histplot(data=df, x="Fare", bins=30, kde=True)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()
```


Fare Distribution
500
400
300
200
100
0
Count
0
100
200
300
400
500
Fare

In [ ]: